Pension Regime Choice under Informality:

Endogenous Sorting between PAYG and Funded Accounts

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Abstract

This paper studies the joint choice of labor-market sector and pension regime in an economy with informality. I build a two-period overlapping-generations model in which heterogeneous workers choose among informal employment, formal employment with a funded account, and formal employment with a pay-as-you-go (PAYG) pension. Funded contributions are portable. PAYG eligibility increases with worker type, non-eligible contributors receive a partial refund, and the pension formula includes a floor and a ceiling. These institutional differences generate two endogenous thresholds without quotas or random-utility shocks. In the benchmark equilibrium, 52.5 percent of workers are informal, 45.4 percent choose formal employment with capitalization, and 2.0 percent choose formal employment with PAYG. Workers below type 0.476 are informal, those between 0.476 and 0.801 choose capitalization, and higher types choose PAYG. Easier PAYG eligibility increases PAYG participation but also raises the consumption tax and informality because the additional benefits are fiscally costly. A local sensitivity exercise preserves positive masses in all three alternatives, although the size of the PAYG group remains small. The results establish a coherent mechanism for regime coexistence; they do not yet constitute an empirical evaluation of a specific pension reform.

Keywords: pension choice; informality; funded accounts; pay-as-you-go; eligibility; overlapping generations. **JEL:** E21; H55; J26; J46.

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1 Introduction

Workers in economies with large informal sectors make two related decisions. They decide whether to participate in formal employment and, when the law permits it, which pension regime will receive their contributions. Treating pension assignment as exogenous suppresses the second margin. It can also conceal a corner solution: if the two formal regimes charge the same contribution rate and promise linear benefits, the regime with the higher implicit return attracts every formal worker.

This paper asks whether informality, funded accounts, and PAYG pensions can coexist as endogenous choices. The model gives each worker three alternatives:

$$\text{informal}, \qquad \text{formal–capitalization}, \qquad \text{formal–PAYG}.$$

The pension regimes differ in three institutional dimensions. A funded account is portable and pays the accumulated market return. PAYG requires the worker to complete a contribution history, represented by an eligibility probability that increases with productivity. A worker who does not qualify receives only a partial refund. Conditional on qualifying, the PAYG pension follows a replacement formula with a floor and a ceiling.

These differences generate comparative advantage across worker types. Low types prefer informality because formal taxes and pension contributions exceed the value of formal-sector benefits. Middle types enter formal employment but choose capitalization because their PAYG eligibility remains low and their account is portable. High types choose PAYG because they are likely to qualify and receive its implicit subsidy. The model therefore produces two endogenous thresholds rather than imposing an assignment rule.

The benchmark equilibrium has the desired three-way sorting. Workers below type 0.476 are informal; workers between 0.476 and 0.801 choose capitalization; and workers above 0.801 choose PAYG. The corresponding population shares are 52.5, 45.4, and 2.0 percent. The consumption tax that balances the consolidated public budget is 16.0 percent.

The small PAYG share is economically informative. PAYG is selected only by high-productivity workers with high eligibility. Moreover, PAYG benefits exceed contemporaneous PAYG contributions in the calibration, so easier eligibility increases fiscal expenditure. When the eligibility intercept rises from -10.00 to -9.60, the PAYG share rises from 1.5 to 2.5 percent, but informality also rises from 49.4 to 55.2 percent and the consumption tax from 13.3 to 18.6 percent. The general-equilibrium fiscal response reverses the partial equilibrium presumption that easier eligibility must increase formality.

The paper makes two contributions. First, it replaces exogenous pension assignment with a joint discrete choice that remains deterministic and interpretable. Second, it shows that eligibility and portability can produce regime coexistence without preference shocks. The present exercise is a mechanism and calibration result. The eligibility profile and the long-period conversion of PAYG benefits are not estimated independently, so the regime shares should not be read as empirical predictions.

The rest of the paper reviews the related literature, presents the model and calibration, reports the equilibrium and sensitivity results, and describes the algorithm.

2 Related literature

The model follows the overlapping-generations approach to social security in Diamond (1977). Mandatory retirement programs can address undersaving (Feldstein, 1985), while present-biased preferences provide a behavioral rationale for mandatory contributions (Laibson, 1998; İmrohoroğlu et al., 2003). Pardo (2023) introduces mandatory retirement saving into an economy with endogenous informality. The present paper adds a second choice margin: formal workers select the institution that receives their contributions.

Pension design cannot be separated from labor informality. Payroll-financed programs alter the value of formal employment (Levy, 2008), and informality reflects choices by workers and firms rather than a residual labor-market state (Ulyssea, 2018). Search frictions and human-capital accumulation create heterogeneous formal careers (Bobba et al., 2021). The shadow economy also changes the information and tax bases available for redistribution (Doligalski and Rojas, 2023).

Quantitative pension models with informality include Joubert (2015) and McKiernan (2020). Related work studies retirement and occupational behavior when formal contributions can be avoided (Ferreira and Parente, 2020; Moreno, 2022). This paper focuses on a narrower mechanism: the interaction between pension eligibility, contribution portability, and the formal–informal margin. The objective is not to impose observed regime shares, but to identify conditions under which the three alternatives are simultaneously optimal for positive masses of workers.

3 Model

3.1 Demography, heterogeneity, and preferences

One model period represents approximately 40 years. Population grows at rate n, labor-augmenting productivity at rate g, and a worker survives to old age with probability $1-m$. Type $i \in [0,1]$ has density

$$f(i) = \frac{2.4450}{1.006341403957648} \exp\left[-\left(\frac{i-0.4655}{0.2329}\right)^2\right], \qquad \int_0^1 f(i)\,\mathrm{d}i = 1. \tag{1}$$

Sector-specific efficiency is

$$z_f(i) = \mathrm{base}_f e^{\rho_f i}, \qquad z_s(i) = \mathrm{base}_s e^{\rho_s i}. \tag{2}$$

Instantaneous utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$. Decision utility and experienced utility are

$$V_j(i) = u(c_{1j}(i)) + \beta(1-m)\delta u(\tilde{c}_{2j}(i)), \tag{3}$$

$$W_j(i) = u(c_{1j}(i)) + (1-m)\delta u(\tilde{c}_{2j}(i)), \tag{4}$$

where $j \in \{I, C, P\}$ denotes informal employment, capitalization, and PAYG. The parameter $\beta < 1$ affects saving and regime choice but is not used as a normative welfare weight.

3.2 Technology and household budgets

Formal and informal output are

$$Y_f = K^\alpha N_f^{1-\alpha}, \qquad Y_s = A_s N_s. \tag{5}$$

Competitive pre-tax prices satisfy

$$\tilde{r} + \delta_k = \alpha(N_f/K)^{1-\alpha}, \quad \tilde{w}_f = (1-\alpha)(K/N_f)^\alpha, \quad \tilde{w}_s = A_s. \tag{6}$$

Taxes map these into the net return r and the effective wages w_f and w_s.

For choice j, voluntary saving $a_j(i) \geq 0$ satisfies

$$(1+\tau_c)c_{1j}(i) + (1+g)a_j(i) = y_{1j}(i), \tag{7}$$

$$(1-m)(1+\tau_c)c_{2j}(i) = (1+r)a_j(i) + p_j(i). \tag{8}$$

Informal income is $y_{1I}(i) = w_s z_s(i)$. Both formal alternatives have

$$y_{1C}(i) = y_{1P}(i) = (1-\tau_\ell-\tau_p)\, w_f z_f(i). \tag{9}$$

Thus the formal regimes differ through retirement benefits, not current contribution rates. Baseline transfers are zero.

3.3 Capitalization

A funded contribution is fully vested and portable:

$$A_C(i) = \tau_p w_f z_f(i), \tag{10}$$

$$p_C(i) = \frac{(1+r)A_C(i)}{1+g}. \tag{11}$$

The contribution adds to the capital stock. The worker receives the accumulated value regardless of whether the subsequent formal career would have satisfied PAYG vesting requirements.

3.4 PAYG eligibility, refund, and pension formula

The probability that type i completes the PAYG eligibility requirement is

$$q(i) = \frac{\exp(\gamma_0+\gamma_1 i)}{1+\exp(\gamma_0+\gamma_1 i)}, \qquad \gamma_1 > 0. \tag{12}$$

The pension conditional on eligibility is

$$\mathcal{B}(y_i) = \min\left\{B_{\max}, \max\left[B_{\min},\, \chi b y_i\right]\right\}, \qquad y_i = w_f z_f(i), \tag{13}$$

where b is the replacement parameter and χ converts the replacement flow into units of the long model period. It also captures the implicit subsidy embedded in the PAYG formula.

If the worker fails to qualify, a fraction φ of the market value of contributions is returned. Expected PAYG retirement income is therefore

$$p_P(i) = q(i)\frac{(1-m)\mathcal{B}(y_i)}{1+g} + [1-q(i)]\varphi\frac{(1+r)\tau_p y_i}{1+g}. \tag{14}$$

PAYG contributions are public revenue and expected benefits, including refunds, are public expenditure.

3.5 Joint choice and endogenous thresholds

Every worker solves the saving problem under all three alternatives and chooses

$$j^*(i) = \arg\max_{j\in\{I,C,P\}} V_j(i). \tag{15}$$

An interior three-regime equilibrium has thresholds $0 < i_1 < i_2 < 1$ satisfying

$$V_I(i_1) = V_C(i_1) > V_P(i_1), \tag{16}$$

$$V_C(i_2) = V_P(i_2) > V_I(i_2). \tag{17}$$

The resulting allocation is

$$j^*(i) = \begin{cases} I, & i < i_1, \\ C, & i_1 \le i < i_2, \\ P, & i \ge i_2. \end{cases} \tag{18}$$

Equation (18) is an equilibrium outcome. The algorithm evaluates all three utilities point by point and does not impose either threshold.

3.6 Aggregation and fiscal closure

Let $d_j(i) = \mathbf{1}\{j^*(i) = j\}$. Regime shares and effective labor are

$$\pi_j = \int_0^1 d_j(i) f(i)\,\mathrm{d}i, \tag{19}$$

$$N_f = \int_0^1 [d_C(i) + d_P(i)] z_f(i) f(i)\,\mathrm{d}i, \tag{20}$$

$$N_s = \int_0^1 d_I(i) z_s(i) f(i)\,\mathrm{d}i. \tag{21}$$

Next period's capital is

$$K' = \frac{1}{1+n}\int_0^1 a_{j^*(i)}(i)f(i)\,\mathrm{d}i + \frac{1}{(1+n)(1+g)}\int_0^1 d_C(i)\tau_p w_f z_f(i)f(i)\,\mathrm{d}i. \tag{22}$$

With PAYG receipts R^{pg} and expected benefits B^{pg}, the consolidated public budget is

$$\tau_c\mathcal{C} + \tau_\ell w_f N_f + \tau_n w_f N_f + \tau_k \widetilde{r} K + R^{pg} = G + \mathcal{T} + B^{pg}. \tag{23}$$

The consumption tax τ_c adjusts to satisfy (23). Consequently, a policy that raises expected PAYG benefits can affect all three choices through the fiscal response.

4 Calibration

Table 1 reports the benchmark. Annual rates are compounded over 40 years: $n = 1.013^{40} - 1$, $g = 1.004^{40} - 1$, and $\beta = 0.97^{40}$. The first block preserves the earlier calibration. The second block governs endogenous regime choice.

Table 1: Baseline parameters

Parameter	Value	Interpretation
m	0.4000	mortality before old age
n	0.6764	population growth per model period
σ	2.5000	CRRA curvature
β	0.2957	behavioral discount factor
g	0.1731	productivity growth per model period
α	0.4000	capital share
A_s	0.1000	informal productivity
ρ_s, ρ_f	2.90, 4.64	type gradients
$\tau_\ell, \tau_n, \tau_p$	0.125, 0.125, 0.125	labor, payroll, pension rates
τ_k	0.3200	capital-income tax
b	0.6400	PAYG replacement parameter
G	0.3300	government purchases
γ_0	-9.7500	eligibility intercept
γ_1	15.0000	eligibility slope
φ	0.5000	refund fraction if not eligible
χ	2.8000	long-period benefit multiplier
$B_{\min}$	0.0500	minimum eligible pension
$B_{\max}$	100.0000	maximum eligible pension

The values of γ_0, γ_1, φ, and χ are not independently estimated. They are selected to implement the institutional mechanism and obtain an interior equilibrium without regime quotas. In the benchmark, $B_{\max}$ does not bind. A complete empirical calibration would discipline eligibility with contribution histories and discipline χ with the present value of statutory PAYG benefits. This limitation matters for the size of the PAYG group and the fiscal effects reported below.

The benchmark is solved in two stages. A 101-type grid produces a stable initial state; a 501-type grid then refines factor prices, thresholds, aggregates, and the consumption tax. The maximum fixed-point residual in the refined equilibrium is 1.6×10^{-7}.

5 Results

5.1 Benchmark equilibrium

Table 2 reports the endogenous steady state. All three choices have positive mass. Informality represents 52.5 percent of workers, capitalization 45.4 percent, and PAYG 2.0 percent. The formal-entry threshold is $i_1 = 0.4763$ and the pension-regime threshold is $i_2 = 0.8005$.

Table 2: Endogenous-choice steady state

Outcome	Value	Outcome	Value
Informal share	0.52525	Capital stock	0.03742
Capitalization share	0.45432	Formal labor	8.74614
PAYG share	0.02044	Informal labor	1.47254
First threshold i_1	0.47626	Consumption tax	0.16007
Second threshold i_2	0.80052	Annualized return	0.05393
Young consumption	0.47367	Formal wage	0.06019
Old-age consumption	0.29448	Voluntary capital	0.00822
Funded capital	0.02921	PAYG contributions	0.00836
Decision welfare	-4.12107	PAYG benefits	0.03570
Experienced welfare	-4.53434	Maximum residual	1.6×10^{-7}

The first threshold reflects the formal tax wedge. Types below 0.476 obtain too little formal-sector income to compensate for labor and pension contributions. Above that threshold, formal productivity grows faster than informal productivity and capitalization becomes optimal. PAYG does not attract middle types because their probability of completing the eligibility requirement remains low.

The second threshold reflects the interaction between eligibility and the implicit PAYG subsidy. At $i = 0.8005$, the expected PAYG benefit overtakes the portable funded balance. Higher types remain in PAYG because eligibility continues to increase and the benefit ceiling does not bind. The two thresholds therefore arise from different comparisons: $V_C - V_I$ at the first threshold and $V_P - V_C$ at the second.

PAYG benefits are large relative to contemporaneous PAYG contributions. The ratio is 4.27 in the benchmark. Part of this difference reflects the model's demographic scaling, and part reflects the multiplier χ required to make PAYG competitive with a funded account over a 40-year period. The consolidated budget finances the difference. This is why regime choice cannot be analyzed independently of fiscal closure.

5.2 Eligibility sensitivity

Table 3 changes only the eligibility intercept. Increasing γ_0 makes every type more likely to qualify while preserving the slope of the profile. Positive masses remain in all three alternatives throughout the reported range.

Table 3: Sensitivity to PAYG eligibility

	$\gamma_0 = -10.00$	Benchmark	$\gamma_0 = -9.60$
Informal share	0.49434	0.52530	0.55183
Capitalization share	0.49057	0.45425	0.42277
PAYG share	0.01510	0.02045	0.02540
Consumption tax	0.13264	0.16012	0.18600
First threshold i_1	0.46353	0.47628	0.48726
Second threshold i_2	0.82018	0.80050	0.78578

Notes: Sensitivity equilibria use 201 types and the refined benchmark as their initial state. All three converge and have maximum fixed-point residuals below 3.6×10^{-6}.

Easier eligibility lowers the second threshold and raises PAYG participation, as expected. However, it also increases expected benefits and the consumption tax. The higher tax raises the first threshold, so informality increases by 5.7 percentage points between the two columns. Capitalization absorbs both movements: some of its workers enter PAYG while others leave formality. The result is a general-equilibrium implication of the calibrated fiscal rule, not a claim that easier eligibility must raise informality in every pension system.

6 Conclusion

Informality, capitalization, and PAYG can coexist without exogenous pension assignment. The model generates two endogenous thresholds because funded balances are portable while PAYG benefits depend on eligibility and include an implicit subsidy. Low types choose informality, middle types choose formal capitalization, and high types choose formal PAYG.

The benchmark produces positive masses in all three alternatives, and local changes in the eligibility intercept preserve the sorting pattern. The exercise also uncovers a fiscal tension. Easier PAYG eligibility raises PAYG participation but increases expected benefits, the consumption tax, and the informal share. Ignoring the budget response would miss this channel.

The quantitative magnitudes remain provisional. Eligibility depends only on productivity; the model does not contain a separate contribution-history state. The benefit multiplier is calibrated rather than estimated, and the PAYG group is small. The next empirical step is to discipline eligibility with observed contribution densities and to value statutory PAYG and funded benefits on a common present-value basis. A second heterogeneity dimension would be appropriate only if the data require workers with similar productivity to choose different regimes. Random-utility shocks are not needed to obtain the reported interior equilibrium.

A Numerical algorithm

A.1 Household alternatives

For each candidate aggregate state (K, N_f, τ_c), the algorithm computes prices and solves three household problems at every diagnostic type:

1. informal employment with no pension contribution;
2. formal employment with a fully funded contribution;
3. formal employment with PAYG eligibility, pension, and refund given by (12)–(14).

The saving solution imposes nonnegative voluntary saving and positive consumption. The algorithm records all three decision utilities and applies (15).

A.2 Locating endogenous thresholds

The diagnostic grid identifies every interval in which the selected alternative changes. Within each interval, a scalar root finder solves the relevant pairwise utility equation. The roots are added to the quadrature nodes. Choice is then evaluated at each interval midpoint and held constant only within that root-bounded interval. This procedure neither assumes the number of crossings nor imposes their order.

A.3 General-equilibrium fixed point

Given the exact choice intervals, trapezoidal quadrature aggregates labor, consumption, voluntary saving, funded contributions, PAYG flows, and welfare. The update map is

$$(K, N_f, N_s, \tau_c) \longmapsto (\widehat{K}, \widehat{N}_f, \widehat{N}_s, \widehat{\tau}_c), \tag{24}$$

where $\widehat{K}$ follows the capital-accumulation equation and $\widehat{\tau}_c$ solves the consolidated budget. Damped updates continue until capital, both labor aggregates, the consumption tax, and the budget residual satisfy tolerance.

The implementation first solves 101 types and uses that state to initialize 501 types. Validation requires positive consumption, nonnegative saving, normalized regime shares, positive masses in all three choices, two interior roots, the sequence $I \to C \to P$, the income identity, and a maximum fixed-point residual below 2×10^{-6}. The sensitivity equilibria repeat the complete fixed point rather than holding prices or taxes fixed.

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